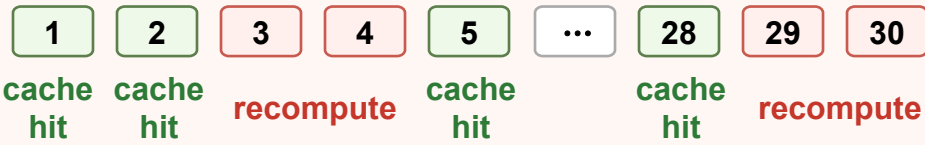


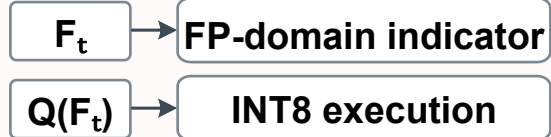
Challenges

1 Decision-Domain Mismatch in Cache+Quant

Illustrative Denoising timeline (30 steps) ■ cache hit ■ recompute



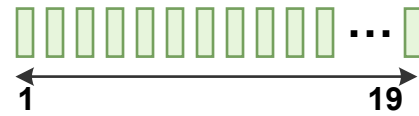
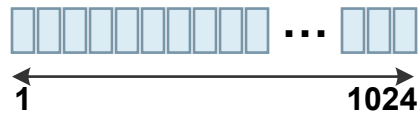
FP indicator $\neq$
INT8-domain
relevance



2 Matrix-Axis Mismatch in Factorized Attention

Spatial: 38 (19x2)
contexts $\times$ 1024-token
rows ($L=1024$)

Temporal: 2048 (1024x2)
contexts $\times$ 19-token rows
($L=19$)



54 $\times$ more contexts

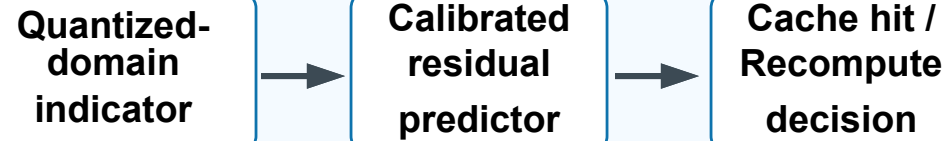
Per head: $d = 72$; $L = 1024$ (spatial), 19 (temporal)

QK^T : $[1, d] \times [d, L] \rightarrow L$ on output axis

PV: $[1, L] \times [L, d] \rightarrow L$ on reduction axis

Proposed Co-Design

1 QAC: Quantization-Aligned Caching



2 DARE: Dimension-Adaptive Engine

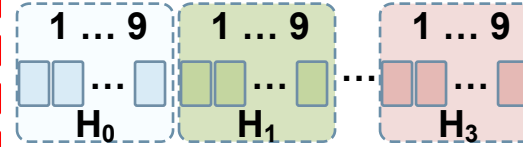
① QK^T output-axis Adaptation (e.g., $T=19$)

Baseline: 1 head $\times$ 36 lanes



19/36=52.8%

Proposed: 4 heads $\times$ 9 lanes



3 tiles/head $\rightarrow$ 70.4%

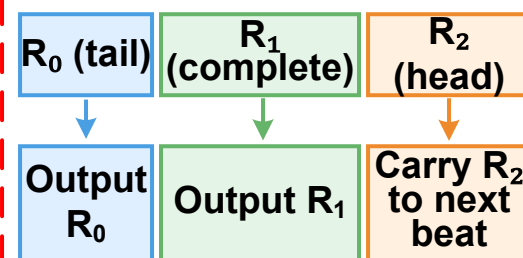
② PV reduction-axis Adaptation (e.g., $T=19$)

Baseline: padding for short rows



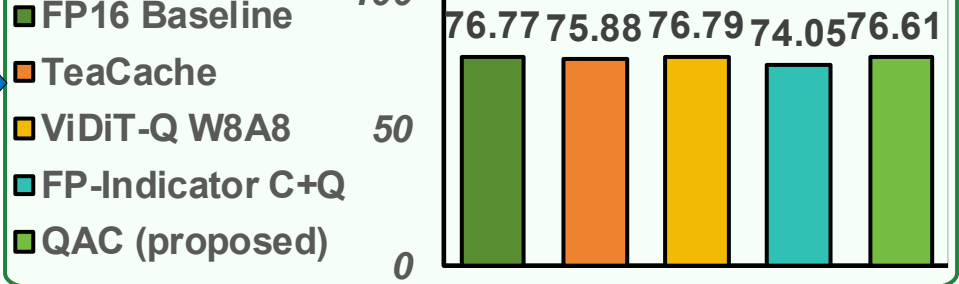
One dot-product result

Proposed: boundary-aware segmented reduction



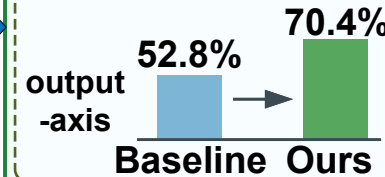
Benefits / Results

1 Algorithm Quality V-Bench (higher is better)

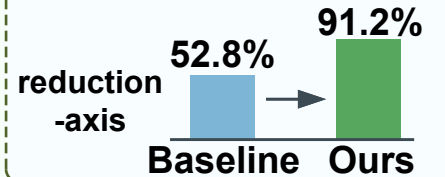


2 Hardware Efficiency

QK^T utilization (Temporal)



PV utilization (Temporal)



3 System-Level Gains

STDiT Step Throughput: 10.96 $\times$ vs A100

Video Throughput: 1.06 $\times$ vs ViDA

Energy Efficiency: 26.59 $\times$ vs A100

Area Efficiency: 2.04 $\times$ vs ViDA